Impact of pharmaceutical care on hospital readmissions for heart failure: a randomised trial

Beatriz Montero-Llorente [1], Covadonga Pérez Menéndez-Conde,[1] Eduardo González Ferrer,[2] Genoveva Teresa López Castellanos,[3] Luis Miguel Bedoya del Olmo,[4] Teresa Bermejo Vicedo[1]

[1]Pharmacy Department, Hospital Universitario Ramón y Cajal, IRYCIS, Madrid, Spain
[2]Cardiology Department, Hospital Universitario Ramón y Cajal, IRYCIS, Madrid, Spain
[3]Internal Medicine Department, Hospital Universitario Ramón y Cajal, IRYCIS, Madrid, Spain
[4]Pharmacology Department, Universidad Complutense de Madrid, Madrid, Spain

Correspondence to
Dr Beatriz Montero-Llorente; bmontero@salud.madrid.org

Received 23 April 2024
Accepted 14 October 2024
Published Online First 27 November 2024

EAHP Statement 4: Clinical Pharmacy Services.

ABSTRACT

Objectives To evaluate the impact of pharmaceutical care on the number of readmissions and visits to the emergency department due to heart failure 30 days after hospital discharge, based on a programme of continuous pharmaceutical care throughout the care process, and to assess the differences between the control and intervention groups at 90 days after discharge (number of readmissions and visits to the emergency department, time from discharge to new readmission or visit to the emergency department).

Methods A single-centre experimental longitudinal prospective open and parallel-group study with balanced randomisation (1:1) was carried out in a tertiary hospital in Spain. Patients with a diagnosis of primary or decompensated heart failure admitted to the Cardiology Service or the Heart Failure and Vascular Risk Unit were recruited between March 2019 and November 2021 and randomly assigned, using a randomised block model, to the control (standard care) or intervention (continuing care model) groups. Epidemiological, clinical and pharmacology data were recorded. As a measure of association, we used the mean difference and the Student's t-test. A p value of <0.05 was considered significant.

Results 296 patients were included (150 randomised to the control group, 146 to the intervention group). The results showed no significant differences between the control and intervention groups in the number of readmissions and visits to the emergency department during the 30 days after discharge (p=0.092), but a statistically significant difference was seen at 90 days (p=0.043). The number of days until the first visit to the emergency department or readmission was higher in the intervention group (p=0.021).

Conclusions Continuous care and follow-up by the pharmacist 30 days after discharge has a neutral impact on hospital readmissions and visits to the emergency department of patients with heart failure, but it is positive in the 90 days following discharge.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Pharmaceutical care has shown multiple benefits in patients with heart failure such as adherence support, symptom control and education on disease management.

⇒ However, these benefits need continued intervention by the pharmacist in order to achieve a reduction in readmissions to hospital and visits to the emergency department.

WHAT THIS STUDY ADDS

⇒ In this study, patients were followed up by the pharmacist for 90 days after discharge, which is a longer period than the 30 days usually described in other studies.

⇒ Furthermore, follow-up telephone calls made up to 3 months after discharge were found to be an appropriate tool to increase the number of days until a new readmission or visit to the emergency department.

HOW MIGHT THIS STUDY AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study could be useful as a model to facilitate the integration of clinical pharmacists in Heart Failure Units.

INTRODUCTION

Heart failure (HF) is a highly prevalent pathology in our society since it is estimated that more than 37 million people worldwide suffer from this disease.[1] It is one of the main causes of hospitalisation in adults and elderly people, with high morbidity and mortality. In Spain, HF causes 3–5% of hospital admissions and is the leading cause of hospitalisation in people aged >65 years. Annual mortality is around 16%, and is close to 60% 10 years after diagnosis.[2 3]

After a first diagnosis of HF, hospital readmissions are frequent, estimated at approximately one readmission per patient per year. In 2020 the percentage of patients with HF who were urgently readmitted within 30 days after hospital discharge due to a process clinically related to their main disease was 12.05%, this being a key indicator of effectiveness and safety.[4] In that year, 15 715 episodes of HF were treated in Madrid hospitals with an in-hospital mortality rate of 8.60%.[4]

Pharmaceutical care (PC) has been defined as 'the professional activity through which the pharmacist links up with the patient (and/or caregiver) and the rest of the health professionals, to care for them according to their needs, proposing strategies to align and achieve short and medium/long-term objectives concerning pharmacotherapy and incorporating new technologies and means available to carry out a continuous interaction with it, to improve health results'.[5] Various studies have shown the benefits of the role of the pharmacist in the care of patients with HF. In a systematic review conducted by Schumacher *et al*, the following actions were recommended to improve morbidity

Check for updates

To cite: Montero-Llorente B, Pérez Menéndez-Conde C, González Ferrer E, *et al*. *Eur J Hosp Pharm* 2025;**32**:126–131.

and mortality: reconciliation of medication and establishment of a medication plan, adherence support and symptom control, and education on HF management.[6]

The therapeutic management of HF includes several lines of treatment, which come together in a complex therapeutic regimen recommended in international guidelines.[7] This circumstance, together with the other comorbidities with which older patients with HF usually present, make polypharmacy a common factor,[8] thus increasing the risk of interactions, adverse effects and lack of adherence to treatment. On the other hand, the transition of patients between the different levels of care constitutes a process of special risk, mainly due to medication errors that can be generated because of the lack of communication and poor coordination between health professionals and the patient.[9 10] The importance of the role of the pharmacist in a safe and continuous transition of pharmacotherapy has been shown by several authors.[11–13] Phatak *et al*[11] concluded that the participation of the pharmacist in the care transition in high-risk patients reduces hospital readmissions or visits to the emergency department (ED) from 39% to 24.8% in the 30 days after discharge. In addition, adverse drug effects or medication errors decreased from 12.8% to 8%.

Different studies have shown that the lack of adherence to treatment is one of the risk factors associated with readmissions for HF.[14 15] Murray *et al*[16] indicated that pharmaceutical intervention can improve adherence and thus decrease healthcare use and costs, but concluded that this benefit probably required continued intervention as the effect diminished considerably when intervention ceased. Odeh *et al*[17] indicated that follow-up calls made by the pharmacist after discharge can be considered a good tool to minimise this problem.

The main objective of this study is to evaluate the impact of PC on the number of readmissions to hospital and visits to the ED for HF in the 30 days after hospital discharge, based on a programme of continuous PC throughout the care process.

The secondary objectives are to assess the differences between the control and intervention groups in the number of readmissions and visits to the ED for HF in the 90 days after hospital discharge, the time elapsed between the two groups in mean hospital stay, time elapsed from discharge to readmissions or ED visits, and to calculate the difference in terms of all-cause mortality in both groups.

METHODS

We carried out a single-centre experimental longitudinal prospective open and parallel-group study with balanced randomisation (1:1) in a tertiary hospital in Spain. The outcome of pharmaceutical intervention in a model of continued PC in patients with HF was compared with those who received standard PC. The study was designated according to the CONSORT statement.

Patients with a diagnosis of primary or decompensated HF, admitted to the Cardiology Service or the Heart Failure and Vascular Risk Unit, were recruited between March 2019 and November 2021. Patients receiving palliative care and those with surgically correctable acute HF were excluded.

From Monday to Friday, the pharmacist reviewed the electronic health record (EHR) to select patients who met the inclusion criteria. For allocation of the participants, a computer-generated list of random numbers was used. They were randomly assigned, using a randomised block model, to the control (standard care) or intervention (continuing care model) group. Block randomisation was by a computer-generated random number list made by a co-investigator. The allocation sequence was concealed from the principal investigator enrolling and assessing participants in sequentially numbered opaque sealed and stapled envelopes. Corresponding envelopes were opened only after the enrolled participants completed all baseline assessments and it was time to allocate the intervention. The pharmacist included 1–2 patients per day in three periods (March–June 2019, September–February 2020 and November 2020–April 2021). The recruitment was greatly affected by the COVID-19 pandemic.

The protocol designed for standard PC in the control group was:

- ► Pharmaceutical reconciliation on admission according to the methodology established in the Guide for the reconciliation of medications in the EDs of the RedFastER Group of the Spanish Society of Hospital Pharmacy.[18]
- ► Validation of medical orders: a daily review of the patient's treatment orders, checking the indication, dose, frequency, route of administration, pharmaceutical form and duration of adequate treatment, as well as possible interactions, duplicities and allergies.
- ► Review of messages between physicians and pharmacists made in the hospital's prescription programme.
- ► Record of pharmaceutical interventions carried out in the prescription programme.

In the intervention group, in addition to the usual PC, the following actions were performed:

- ► Recommendations, according to the clinical judgement of the pharmacist, on the pharmacotherapeutic treatment of HF prescribed by the physician (verifying the appropriateness of the medication, dosage form and route of administration, adjusting medication dosages to optimise efficacy and safety, identifying and managing potential drug interactions, monitoring for adverse drug reactions, promoting patient adherence to the treatment plan) and their registration in the EHR for subsequent evaluation. The recommendations also included de-prescribing in patients polymedicated and with multimorbidity, and the detection of drug-related problems and negative outcomes associated with medication classified according to the Third Consensus of Granada. In patients aged >65 years with multiple pathologies, the STOPP/START criteria were used.[19]
- ► Pharmaceutical reconciliation at discharge based on the RedFastER methodology[18] and patient information. The patients received a pharmacotherapeutic report prepared using the InfoWin programme, a tool that provides information on the correct use of medications through understandable and visual language.

The information was collected through an initial interview with the patient and/or caregiver, reports from the HORUS platform (the Health Service information system of Madrid that allows the visualisation of clinical information on patients) and the MUP (a tool included in the Health Service information system that provides a unique and complete pharmacotherapeutic history of the patient), reports collected in the EHR, telephone calls to patients and reports from the residence if needed.

The following variables were recorded: sex, age, number, therapeutic group and active principal of the drugs they were taking at the time of the inclusion, number of previous pathologies, type of HF and medication reconciliation errors detected on admission. To assess the primary endpoint, we recorded the number of readmissions to the hospital or visits to the ED for HF within 30 days of discharge. The variables to evaluate the secondary outcomes were the number of hospital readmissions or ED visits within 90 days of discharge, the number of days of stay and days until readmissions or visits to the ED, the number of deaths and their causes.

Follow-up in the control group consisted of making a telephone call in months 1 and 3 after hospital discharge to find out if there had been any readmission or visit to the ED. In the intervention group, a telephone call was also made 3 and 14 days after discharge to reinforce the information and provide PC support if necessary.

To calculate the sample size we used the results published in the work by Phatak *et al*[11] and data from the Observatory of the Public Health Service in Madrid.[4] To reduce the rate of hospital readmissions or ED visits within 30 days of discharge from 20% to 10% with a two-sided 5% significance level and a power of 70%, a sample size of 157 patients per group was necessary.

Statistical analysis was performed using the SPSS programme (Statistical Package for Social Sciences). A first general review was carried out using frequency distribution for categorical variables and determination of minimums, maximums and presence of missing for the quantitative variables, with the subsequent verification of the discordant data and the correction of the errors detected. A descriptive analysis of the characteristics of the patients concerning the explanatory variables was performed. If the distribution of the variables was considered symmetrical (verified using graphic methods and the Kolmogorov–Smirnov normality test), the mean±SD was determined. In the case of asymmetrical distributions, the median plus the 25th and 75th percentiles was determined. For the discrete variables, absolute and relative frequencies were determined. As a measure of association of the main variable, we used the mean difference and the Student's t-test after previously checking the normal distribution of the sample. A p value of <0.05 was considered significant.

The study was approved by the Research Ethics Committee with Medicines. The treatment of the data was carried out according to the European Regulation of Data Protection 2016/679.

RESULTS

A total of 296 patients were included (159 (53.7%) women, mean±SD age 81.7±6.8 years); 150 patients were randomised to the control group and 146 to the intervention group. The primary analysis was intention-to-treat and involved all patients who were randomly assigned. The patients were followed up until 3 months after discharge. The demographic characteristics of the two groups are summarised in table 1. Hypertension (78.7%), atrial fibrillation (62.5%), dyslipidaemia (54.5%) and diabetes (42.4%) were the most frequent comorbities.

Table 1 Demographic characteristics of the two study groups

	Control group (n=150) N (%)	Intervention group (n=146) N (%)
Sex		
Men	67 (44.7)	70 (47.9)
Women	83 (55.3)	76 (52.1)
Age (years)	81.6±6.9	81.9±6.7
Type of HF		
Systolic	56 (37.3)	47 (32.2)
Diastolic	94 (62.7)	99 (67.8)
Current medications		
<5 drugs	6 (4)	3 (2.1)
≥5 drugs	144 (96)	143 (97.9)
Previous pathologies	6.2±2.5	6.4±2.5

HF, heart failure.

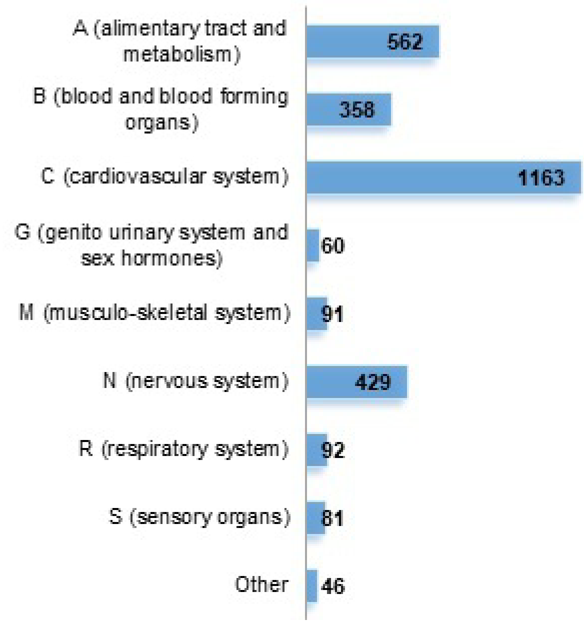

Figure 1 Number of prescriptions by pharmacological group according to the ATC classification.

The number of medications being taken by the patients at the time of inclusion was 9.6±3.1, with 287 patients (96.9%) taking ≥5 medications. The distribution of the total of prescriptions (2882 medications) by pharmacological group is shown in figure 1, with those in group C being the most prescribed according to the ATC classification (40.3%), followed by group A (19.5%), group N (14.9%) and group B (12.4%).

Focusing on the cardiovascular system, furosemide was the most frequently prescribed drug at the time of admission (n=192), followed by bisoprolol (n=113), simvastatin (n=64) and atorvastatin (n=56). The distribution by pharmacological subgroups of the active principles within this group is shown in figure 2.

At the time of inclusion, reconciliation treatment was performed in all patients. We found 253 medication-related events (MREs), most of which were omission errors followed by dosage errors and commission errors (table 2). None of the MREs increased the hospitalisation time, caused permanent damage or resulted in a fatal outcome. Of the omission errors that involved medications from group C, statins (n=14) and angiotensin-converting enzyme inhibitors/angiotensin receptor blockers (n=11) were the most commonly implicated.

The type and grade of acceptance of the recommendations made by the pharmacist to the physicians in the intervention group are shown in table 3.

Information was given at discharge to 95.9% of the patients in the intervention group (six patients were discharged before being able to be given the pharmacotherapeutic report). Telephone follow-up was performed in 91.2% of the patients (93.3% in the control group and 89% in the intervention group).

Twenty five (17.1%) patients in the intervention group visited the ED or were readmitted 30 days after hospital discharge compared with 30 (20%) in the control group (p=0.092). Three months after hospital discharge, 28 (19.2%) patients in the intervention group visited the ED or were readmitted compared with 37 (24.7%) in the control group. This difference was statistically significant (p=0.043). The results are shown in table 4.

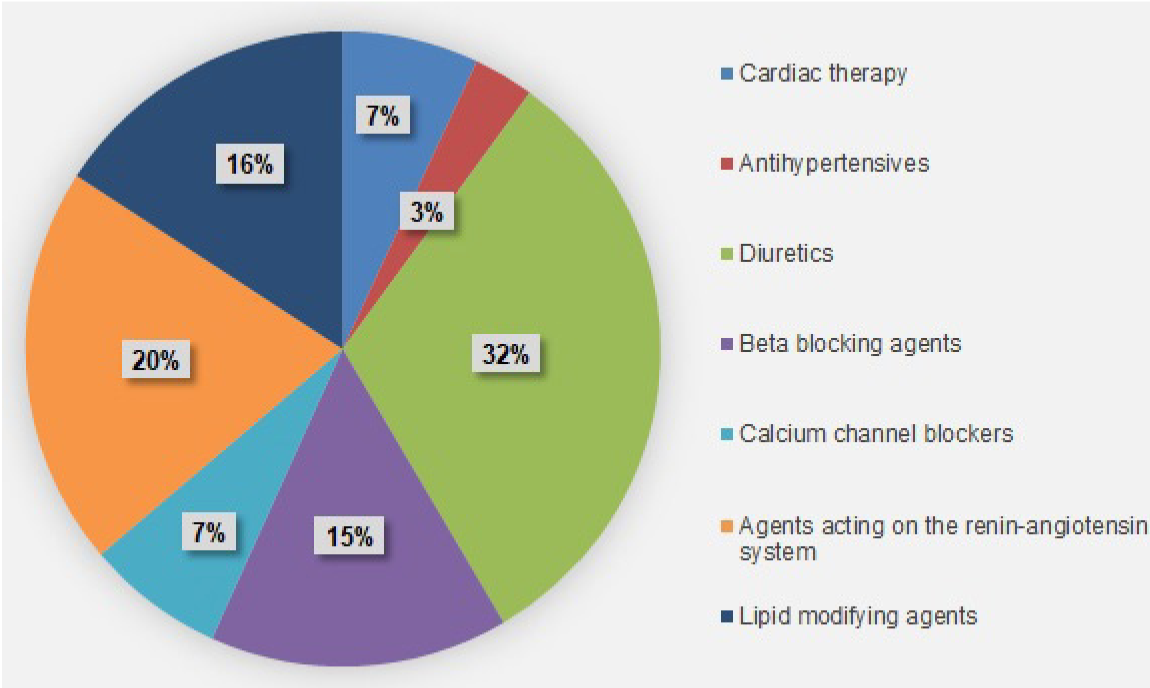

Figure 2 Distribution of prescriptions by subgroup within group C of the ATC classification.

The number of days until the first visit to the ED or readmission after hospital discharge was higher in the intervention group (46.8 days vs 28.2 days, p=0.021); this difference was statistically significant. The difference in the mean length of stay was 0.5 days between the two groups (p=0.52).

Twelve patients in the intervention group died compared with eight in the control group (p=0.639).

DISCUSSION

The main objective of our study is to evaluate the impact of the pharmaceutical intervention on the number of readmissions and visits to the ED for HF in the 30 days after discharge. A slightly lower percentage occurred in the intervention group (17.1% vs 20%) but the difference was not statistically significant. In this regard, Warden *et al*[20] obtained a statistically significant difference in the number of readmissions at 30 days for any cause (17% vs 38%), which became non-significant when exclusively evaluating readmissions due to HF (6% vs 18%). Moye *et al*[21] also obtained a statistically significant result when assessing any cause of readmission (12% vs 25%), but this was not significant when analysing readmissions due to HF (55% vs 58%). In both studies the pharmacist made two telephone calls after hospital discharge at 14 and 30 days and, in addition to the information about the treatment, its regimen and its possible adverse effects, they monitored the patient's weight and fluid intake records as well as providing advice on lifestyles and other recommendations that could influence the results obtained.

The percentage of readmissions for HF at 30 days obtained in the study conducted in South Florida by Salas and Miyares[22] was 16.6%. This result is similar to ours, but with a substantially smaller sample size. Neu *et al*[23] showed in a study carried out in Detroit that the difference in the number of readmissions due to HF was statistically significant (10.5% vs 17.3%), which can be explained by the fact that post-discharge follow-up was also carried out by doctors and nurses as well as by the pharmacist, and also considering the significantly larger sample size than in other studies.

The variable of readmissions and visits to the ED for HF 90 days after hospital discharge has not usually been measured in other studies of patients with HF, which shows that it would be interesting to investigate the research on the value of more continuous PC over time. As the intervention was implemented for both sexes, all ages and all types of HF, the results indicate

Table 2 Classification of medication-related events detected at the time of inclusion

	Control group (n=127) N (%)	Intervention group (n=126) N (%)
Omission	59 (46.4)	67 (53.2)
Dosage	32 (25.2)	27 (21.4)
Commission	19 (14.9)	12 (9.5)
Frequency	11 (8.7)	14 (11.1)
Incorrect medication	3 (2.4)	3 (2.4)
Incomplete prescription	1 (0.8)	3 (2.4)
Route	2 (1.6)	0 (0)

Table 3 Grade of acceptance of the interventions made by the pharmacist, classified according to the Third Consensus of Granada

	Interventions made by the pharmacist (n=1095) n (%)	Accepted interventions (n=627) n (%)
Untreated health problem	215 (19.6)	174 (27.7)
Effect of unnecessary medicine	272 (24.9)	208 (33.2)
Non-quantitative ineffectiveness	157 (14.3)	38 (6.1)
Quantitative ineffectiveness	141 (12.9)	32 (5.1)
Non-quantitative safety problem	147 (13.4)	80 (12.8)
Quantitative safety problem	163 (14.9)	95 (15.1)

Table 4 Follow-up variables after hospital discharge

	Control group (n=150) N (%)	Intervention group (n=146) N (%)	P value
Visits to ED			
In the 30 days after discharge	7 (4.7)	8 (5.5)	0.716
In the 90 days after discharge	11 (7.3)	10 (6.8)	0.645
Readmissions			
In the 30 days after discharge	23 (15.3)	17 (11.6)	0.261
In the 90 days after discharge	26 (17.3)	18 (12.3)	0.157
Visits to ED and readmissions			
In the 30 days after discharge	30 (20)	25 (17.1)	0.092
In the 90 days after discharge	37 (24.7)	28 (19.2)	0.043
Mean length of stay (days)	3.8	3.3	0.524
Time until visit to ED or readmission (days)	28.2	46.8	0.021
Mortality	8 (5.3)	12 (8.2)	0.639

ED, emergency department.

that all patients with HF would benefit from PC after discharge. Bae-Shaaw *et al*[24] evaluated the effect of the pharmaceutical intervention on readmissions for any cause 30 and 90 days after discharge in northern California, including patients admitted with a diagnosis of acute myocardial infarction, chronic obstructive pulmonary disease, HF and pneumonia. When analysing the subgroup of patients with HF, they found statistically significant differences at 90 days but not at 30 days, as in our study.

López Cabezas *et al* showed in a study in Spain that an educational intervention at hospital discharge in patients with HF, carried out by a pharmacist in coordination with the rest of the team, reduces hospital readmissions and total days of hospitalisation, improving adherence to treatment without increasing the costs of care.[25] Our research shows that, although the difference in the length stay in the two groups was not statistically significant (3.8 vs 3.3 days), the number of days until readmission or ED visit was significant (28.2 vs 46.8 days). This may be due to the information received at discharge through a detailed pharmacotherapeutic report, as well as follow-up telephone calls in which adherence to treatment was reinforced and doubts were resolved in this regard. The number of days until readmission or visit to the ED was also measured by Neu *et al*,[23] with a statistically significant result ($p=0.012$), which shows the importance of the role of the pharmacist in the follow-up after the hospital discharge. However, in other studies such as that by Bae-Shaaw *et al*,[24] no statistically significant differences were found when measuring this variable.

Compared with other pharmacist interventions published,[22 26] our study focuses on a more comprehensive approach, combining direct patient education, post-discharge follow-up and ongoing support for both adherence and optimisation of treatment plans. This integrated model reflects an alignment with recent findings which suggests that multidisciplinary approaches, including pharmacists, are more effective in managing chronic conditions like HF.

Strengths and limitations

The main limitation of our study is that there was no pharmacist on site in the hospitalisation area at the time of discharge, and several patients in the intervention group could not receive written information at this time. Likewise, since most of the patients were older, telephone follow-up was sometimes difficult.

The main strength of our research, in addition to the large sample size, is the intervention of the pharmacist throughout all of the care process and also our follow-up period was longer than that of most of the studies published.

Our next steps involve conducting further studies with a longer follow-up period to validate these results and explore additional benefits of continuous PC. We also aim to investigate the cost-effectiveness of this intervention to support its broader implementation within the healthcare system.

CONCLUSION

Our study shows that continuous care and follow-up by the pharmacist in the 30 days after discharge has a neutral impact on hospital readmissions and visits to the ED of patients with HF, but at 90 days after discharge it has a positive impact. These results highlight the importance of integrating the pharmacist into the healthcare team caring for these patients.

Contributors BML: Guarantor. Study conception, design, acquisition of data, interpretation and analysis, writing. CPM-C: study conception, design, interpretation and analysis, review and finalisation of the manuscript. EGF: review and finalisation of the manuscript. GTLC: review and finalisation of the manuscript. LMBO: review and finalisation of the manuscript. TB-V: study conception, design, interpretation and analysis, review and finalisation of the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Consent obtained directly from patient(s)

Ethics approval This study involves human participants and was approved by the Research Ethics Committee with Medicines (Hospital Universitario Ramón y Cajal) ID: 055/16. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement No data are available.

ORCID iD

Beatriz Montero-Llorente http://orcid.org/0000-0002-9909-665X

REFERENCES

1 Ziaeian B, Fonarow GC. Epidemiology and aetiology of heart failure. *Nat Rev Cardiol* 2016;13:368–78.
2 Sayago-Silva I, García-López F, Segovia-Cubero J. Epidemiología de la insuficiencia cardiaca en España en los últimos 20 años. *Rev Esp Cardiol* 2013;66:649–56.
3 Montes-Santiago J, Arévalo Lorido JC, Cerqueiro González JM. Epidemiology of acute heart failure. *Med Clin (Barc)* 2014;142 Suppl 1:3–8.
4 Observatorio de resultados del Servicio Madrileño de Salud. Consejería de Sanidad. Available: http://observatorioresultados.sanidadmadrid.org/HospitalesFicha.aspx?ID=30 [Accessed 18 Apr 2022].
5 Morillo-Verdugo R, Calleja-Hernández MÁ, Robustillo-Cortés M de LA, *et al*. A new definition and refocus of pharmaceutical care: the Barbate Document. *Farm Hosp* 2020;44:158–62.
6 Schumacher PM, Becker N, Tsuyuki RT, *et al*. The evidence for pharmacist care in outpatients with heart failure: a systematic review and meta-analysis. *ESC Heart Fail* 2021;8:3566–76.
7 McDonagh TA, Metra M, Adamo M, *et al*. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;42:3599–726.
8 Mastromarino V, Casenghi M, Testa M, *et al*. Polypharmacy in heart failure patients. *Curr Heart Fail Rep* 2014;11:212–9.
9 Kripalani S, LeFevre F, Phillips CO, *et al*. Deficits in communication and information transfer between hospital-based and primary care physicians: implications for patient safety and continuity of care. *JAMA* 2007;297:831–41.
10 Magallón Martínez A, Pinilla Rello A, Casajús Lagranja P, *et al*. Pharmaceutical care for the patients admitted to a multidisciplinary complex chronic patient unit. *Farm Hosp* 2023;47:T106–12.
11 Phatak A, Prusi R, Ward B, *et al*. Impact of pharmacist involvement in the transitional care of high-risk patients through medication reconciliation, medication education, and postdischarge call-backs (IPITCH Study). *J Hosp Med* 2016;11:39–44.

12 Anderson SL, Marrs JC. A Review of the Role of the Pharmacist in Heart Failure Transition of Care. *Adv Ther* 2018;35:311–23.
13 Ruiz Ramos J, Calderón Hernanz B. [Translated article] Pharmaceutical care in emergency department. *Farm Hosp* 2023;47:T97–9.
14 Al-Tamimi MA-A, Gillani SW, Abd Alhakam ME, *et al*. Factors Associated With Hospital Readmission of Heart Failure Patients. *Front Pharmacol* 2021;12:732760.
15 Tun H. Importance of medication compliance and lifestyle modification in heart failure readmission: single centre cohort study. *Eur J Prev Cardiol* 2021;28.
16 Murray MD, Young J, Hoke S, *et al*. Pharmacist intervention to improve medication adherence in heart failure: a randomized trial. *Ann Intern Med* 2007;146:714–25.
17 Odeh M, Scullin C, Fleming G, *et al*. Ensuring continuity of patient care across the healthcare interface: Telephone follow-up post-hospitalization. *Br J Clin Pharmacol* 2019;85:616–25.
18 Calderón Hernanz B, Oliver Noguera A, Tomás Vecina S, *et al*. Guía para la Conciliación de los medicamentos en los servicios de urgencias. *Emergencias* 2013;25:204–17.
19 Delgado Silveira E, Montero Errasquín B, Muñoz García M, *et al*. Mejorando la prescripción de medicamentos en las personas mayores: una nueva edición de los criterios STOPP-START. *Rev Esp Geriatr Gerontol* 2015;50:89–96.
20 Warden BA, Freels JP, Furuno JP, *et al*. Pharmacy-managed program for providing education and discharge instructions for patients with heart failure. *Am J Health Syst Pharm* 2014;71:134–9.
21 Moye PM, Chu PS, Pounds T, *et al*. Impact of a pharmacy team-led intervention program on the readmission rate of elderly patients with heart failure. *Am J Health Syst Pharm* 2018;75:183–90.
22 Salas CM, Miyares MA. Implementing a pharmacy resident run transition of care service for heart failure patients: Effect on readmission rates. *Am J Health Syst Pharm* 2015;72:S43–7.
23 Neu R, Leonard MA, Dehoorne ML, *et al*. Impact of Pharmacist Involvement in Heart Failure Transition of Care. *Ann Pharmacother* 2020;54:239–46.
24 Bae-Shaaw YH, Eom H, Chun RF, *et al*. Real-world evidence on impact of a pharmacist-led transitional care program on 30- and 90-day readmissions after acute care episodes. *Am J Health Syst Pharm* 2020;77:535–45.
25 López Cabezas C, Falces Salvador C, Cubí Quadrada D, *et al*. Randomized clinical trial of a postdischarge pharmaceutical care program vs regular follow-up in patients with heart failure. *Farm Hosp* 2006;30:328–42.
26 Milfred-LaForest SK, Gee JA, Pugacz AM, *et al*. Heart Failure Transitions of Care: A Pharmacist-Led Post-Discharge Pilot Experience. *Prog Cardiovasc Dis* 2017;60:249–58.